ANNet: Asymmetric Nested Network for Real-Time Cloud Detection in Remote Sensing

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***Abstract*— Cloud detection is one of the crucial tasks in the field of remote sensing, which is regarded as binary classification at the pixel level. Although a large number of recent deep learning-based methods have made great progress, most of their appealing performances come at the expense of a large amount of computation, which reduces the real-time performance accordingly. In order to bridge the gap between segmentation performance and inference speed, we propose a novel asymmetric nested network (ANNet) architecture termed ANNet, which is designed for real-time cloud detection with excellent performance. In the encoder branch of ANNet, we introduce an effective tiny U-shape block (TUB) to enrich detailed spatial contexts in each stage, which also allows ANNet to embed spatial recovery ability earlier, and a lightweight, simple feature fusing module (SFFM) is designed to refine the semantic features map at a lower level of TUB for better performance. Following the encoder, which is diverse from the most symmetric U-shape approaches, an asymmetric and lightweight decoder (ALD) with only convolution and bilinear up-sample operations is employed for spatial recovery. We also, moreover, demonstrated that using a constant channel size instead of a larger channel volume as the network goes deeper is an efficient and effective design for cloud detection tasks. Substantial experiments are performed on GF1_WHU and 95-Cloud datasets, which show that ANNet has achieved excellent performance with low computation cost compared to most of the existing state-of-the-art methods. On GF1_WHU dataset, ANNet-l achieves 93.79% on mIoU at 125 FPS, while ANNet-s with only 29.5 K parameters yields 90.74% mIoU at 251 FPS on the Nvidia RTX 2080Ti.**

***Index Terms*— Cloud detection, nested encoder (NE), real-time semantic segmentation, remote sensing.**

I. Introduction

With the rapid development of remote sensing technology, remote sensing images are widely used in various fields, such as military information analysis, crop production forecasting, environmental monitoring, mineral resource surveying, and geographic information integration. In recent years, a growing number of remote sensing satellites have been launched, and the spatial and temporal resolution of remote sensing images has been increasing, which has resulted in an exponential increase in the data volume of remote sensing images. According to [1], the global annual mean cloud cover is, however, approximately 67%. As a result, clouds significantly challenge target detection, object recognition, and other tasks and ultimately cause incorrect spectral information extraction. Cloud detection is, therefore, a vital step in remote sensing imagery processing.

Traditional cloud detection algorithms typically employ manually designed rules and expert knowledge to conduct threshold analysis based on the cloud's structure, texture, and other physical properties, as well as spectral characteristics. Typical application scenarios include the AVHRR processing scheme over clouds, land, and ocean (APOLLO) [2], the clouds from the advanced very high-resolution radiometer (CLAVRS) [3] and the Fmask [4]. While traditional methods were straightforward, their classification performance proved unsatisfactory in practical applications when dealing with complex ground scenes.

As machine learning methods come into the spotlight, methods like the support vector machine (SVM) [5], random forest [6], Bayesian spatial–temporal random field [7], and MAP–Markov random field (MRF) framework [8] are adapted to address the cloud coverage issues and have made great progress. But there is still a lot of room for improvement in detection accuracy and efficiency, since most of them mainly focus only on the shallow textural features in optical satellite remote sensing imagery.

Recently, deep learning promoted further progress in cloud detection, and region-based approaches like superpixel segmentation [9] are the most representative. The essence of such methods lies in using simple linear iterative clustering (SLIC) or its variant schemes to transform homogeneous pixels into superpixels, and it has been found that processing each superpixel as a unit is significantly more manageable. Conditional random fields (CRFs) [10] and MRFs [11] are, however, used to refine the final masks, which is time-consuming.

Deep learning-based semantic segmentation techniques have also seen significant advancements. FCN [12], SegNet [13], PSPnet [14], Deeplab [15] along with its variants [16], [17], and UNet [18] along with its variants, have made important strides in the field of natural image semantic segmentation. Different from the natural images, the shapes of clouds in

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optical remote sensing imagery, however, exhibit randomness, diversity, and significant scale variations, which renders cloud segmentation more challenging, especially in the following two aspects.

First, the challenge is to effectively extract and aggregate multiscale features in the field of cloud detection. Because the shallower convolutional layers contain detail contexts instead of global properties, deeper layers provide higher-level semantic information but lack details. As the network deepens, textural information from shallow layers gradually abstracts into semantic context, which results in the shortage of detailed contexts in the final map for classification; thus, to effectively extract and aggregate the shallow detailed and semantic context features is one of the key issues. To this point, the DeepLab series approach [17], [19], [20] employs residual connections and atrous convolution to capture contextual information, and ASPP are proposed to refine both detailed and semantic features. Dev et al. [21], Shi et al. [22], and Guo et al. [23] draw inspiration from the symmetric encoder-decoder structure of U-Net, improving the skip connections to alleviate the loss of detailed information during decoder recovery. Zhang et al. [24], He et al. [25], Zhao et al. [26], and Wu et al. [27] introduce extra supervision to enhance the segmentation accuracy of cloud boundaries. To further pursue segmentation accuracy, Transformer [28], [29] and the combinations of CNN and Transformer [30], [31] are involved, although the model sizes and computational demands soar. In addition, various attention mechanisms are proposed to help the network to focus on the critical features for cloud detection, which also further increases the computational loads.

Second, the challenge to lighten the model. In many application scenarios, especially on mobile or edge devices where computational resources are limited. Lightweight models ensure acceptable inference speed and power consumption, which is particularly important for real-time scenarios. To address this issue, Li et al. [32] use a lighter MobileNet as the backbone network instead of ResNet. Guo et al. [23], Chen et al. [33], and Zhang et al. [34] lighten the overall network by designing more efficient attention mechanism modules. Li et al. [32] and Luo et al. [35] designed a lightweight feature fusion module to balance performance and efficiency. All these methods, however, make the model more lightweight through a partial replacement approach. They do not fundamentally redesign the model for the task of cloud detection, and thus, fail to meet the requirements for real-time processing.

High-performance and lightweight design, therefore, seem to be mutually exclusive in existed cloud detection approaches. Because of most of the high-performance methods tend to have a large number of parameters and high computational requirements while lightweight models often fall short in terms of detection performance.

In this article, to address this issue, we propose a high-performance, lightweight, and real-time model termed as asymmetric nested network (ANNet) for cloud detection in remote sensing, as shown in Fig. 2. Specifically, inspired by [36], we propose a nested encoder (NE) with tiny U-shape blocks (TUBs) to efficiently extract multiscale features to improve the overall performance, followed by a lightweight simple feature fusing module (SFFM) to efficiently refine the semantic features map at the bottom of the encoder for better performance, and finally an asymmetric and lightweight decoder (ALD) is designed to parse high-level semantic information into final segmentation result effectively. As illustrated in Fig. 1, our network ANNet-s has only 29.5 K parameters and achieves 90.74% mIoU at 251 FPS on the GF1_WHU test set. The main contributions of this study are as follows.

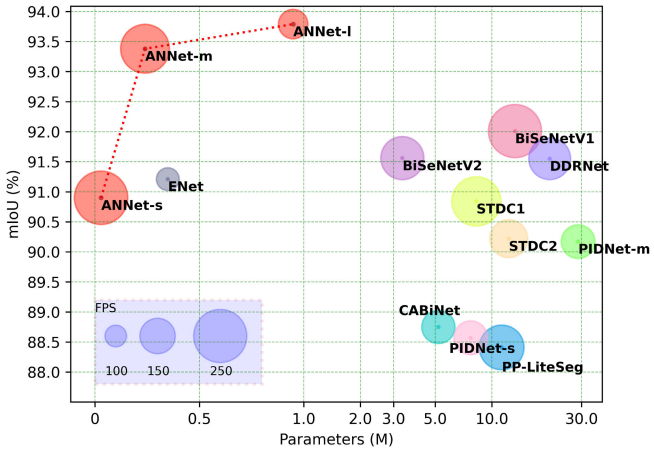


Fig. 1. Comparisons between real-time semantic segmentation networks and the ANNet on GF1_WHU dataset.

1) We propose a NE with TUBs. A TUB module can effectively extract features at the corresponding scale, and the concatenation of TUB modules facilitates better integration of multiscale information.
2) We propose an ALD, which mitigates the redundancy of the decoder while maintaining high decoding accuracy.
3) Based on the above modules, we propose ANNet, a novel real-time semantic segmentation model for cloud detection in remote sensing. Extensive experiments demonstrate our SOTA performance in terms of both accuracy and speed.

II. Related Work

A. High-Accuracy U-Shape Architecture

1) CNN-Based Method: Early semantic segmentation methods were largely based on the encoder-decoder architecture. Since the advent of U-Net in 2015, numerous networks have built upon it as a fundamental framework, introducing various improvements. U-Net++ [37] introduces multiple skip connections between each encoder and decoder layer, dividing these connections into different scales, which enables the model to better capture feature information at various scales. Attention U-Net [38] introduces attention modules for automatic learning of important regions and assigns higher weights to them. This allows the model to focus more on critical semantic information. U2-Net [36] is a nested U-shaped structure. Residual U-blocks (RSUs) blend receptive fields of different sizes, enabling U2-Net to capture more contextual

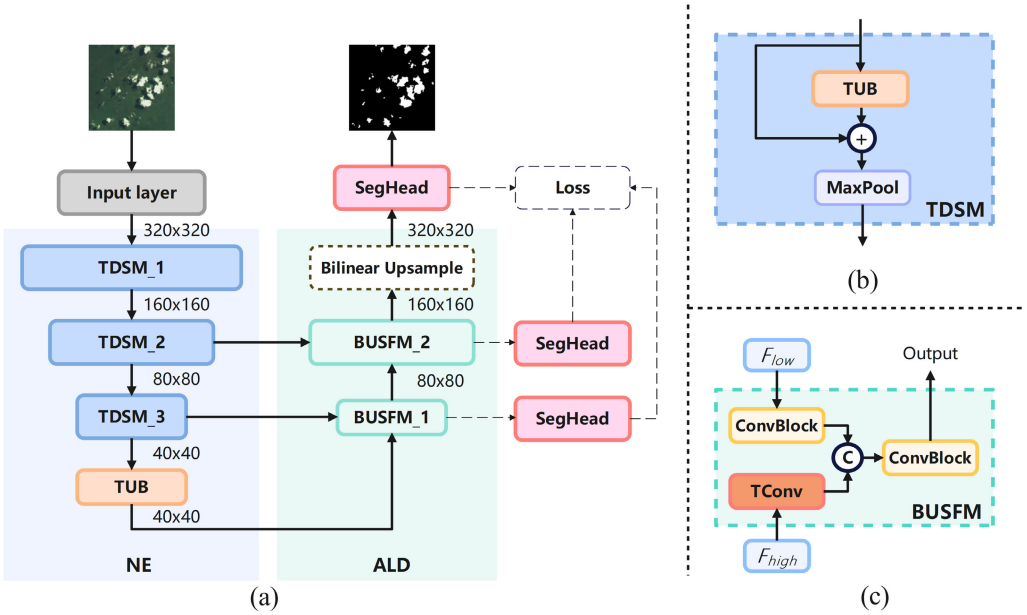


Fig. 2. Architecture of the ANNet. (a) Details of ANNet with the blue area for NE and green area for ALD. (b) TDSM. (c) BUSFM.

information from various scales. CloudU-Net [22] uses the UNet as its backbone for feature extraction and fusion, and it employs CRF to optimize the results of cloud detection. CFCA-Net [34] employs a cascaded feature attention mechanism to boost the network's focus on crucial color and texture features of the cloud. SCUCNet [39] and CDANet [40], respectively, use Deeplab and U-Net as their backbone networks for shadow detection.

2) Transformer-Based Method: The transformer was initially applied in the field of natural language processing (NLP) until the appearance of the paper [41], which successfully introduced the transformer architecture into the domain of computer vision. This article presents the vision transformer (ViT), innovatively partitioning images into fixed-sized patches and treating them as sequential data to be processed by a transformer, thereby enabling large-scale application of transformers in visual recognition tasks. To address the issues in ViT stemming from its low-resolution feature maps and quadratic complexity growth with image size, rendering the structure ill-suited for dense visual tasks or serving as a backbone for high-resolution input images, the Swin transformer [42] architecture is proposed, offering an optimal trade-off between accuracy and speed. Swin-UNet [43] combines the advantage of transformer in feature extraction with the architectural advantage of UNet. CSDFormer [28] employs a hierarchical transformer structure in the encoder stage. Each transformer layer contains several multihead self-attention mechanisms for calculating pixel-wise long-distance connectivity in cloud detection.

3) Hybrid Method: To fully leverage the advantages of both CNN and transformer architectures, several papers have begun proposing new networks that integrate CNNs and transformers into a single, cohesive structure. TransUnet [44] ingeniously interweaves CNN and transformer architectures, preserving the inherent strengths of CNN structures for local feature extraction and spatial invariance while incorporating the transformer's potent capability for global information modeling. Transfuse [45] parallels a CNN encoder with a transformer encoder, subsequently integrating multilevel features generated from both the CNN and transformer branches via a feature fusion module. The UNeXt [46] network improves detection efficiency by combining CNN modules with tokenized MLP modules. DA-TransUnet [47], building upon TransUnet, incorporates a dual attention mechanism over spatial and channel dimensions to enhance feature representation capabilities. By adopting a convolution-transformer hybrid structure network, HyCloudX [30] integrates the advantages of both convolution and transformer architectures to synergize convolution's local detailed features and transformer's global semantic information for clouds and shadows.

B. Real-Time Semantic Segmentation

1) Lightweight Encoder-Decoder Network: ENet [48] employs early downsampling strategies and dilated convolutions to reduce computational costs associated with processing large images and feature maps. ICNet [49] balances computational efficiency and segmentation quality by cascading subnetworks of different resolutions, effectively fusing feature information across various scales. SwiftNet [50] employs depthwise separable convolutions to reduce computational complexity and incorporate attention mechanisms. These mechanisms assign distinct weights during the fusion of features from different layers, enabling the model to adaptively adjust the importance of features according to the content of the input image. DFANet [51] employs a lightweight encoder constructed using depthwise separable convolutions to maintain a small computational complexity. SFNet [52] proposes a Flow Alignment Module, a method for learning semantic flow between feature maps at adjacent scales. CloudSegNet [21]

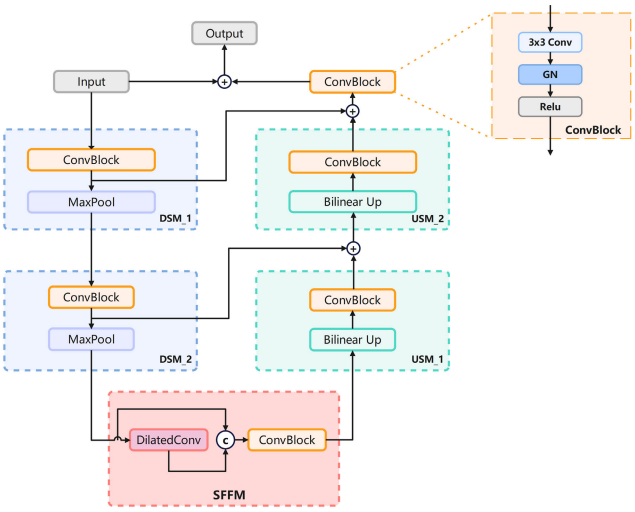


Fig. 3. Structure of TUB ($n = 2$). The red dashed box shows the SFFM. The yellow dashed box shows the ConvBlock.

adopts a streamlined SegNet as its backbone, making the cloud detection model sufficiently lightweight.

2) Dual-Branch Network: BiSeNet v1 [53] and BiSeNet v2 [54] employ a dual-branch network architecture along with a feature fusion module, using complementary detail and semantic branches to enhance operational efficiency. CABiNet [55] introduces a lightweight context aggregation module for aggregating multiscale features. STDCSeg [56] proposes the STDC module with reduced channel receptive field enlargement and devises an efficient single-path backbone capable of strengthening feature representation at a lower computational cost. In order to eliminate redundancy in dual-branch networks, STDCSeg guides features using detailed ground truths, further enhancing efficiency. The PP-LiteSeg [57] network achieves a lightweight design by focusing on the decoder, multiscale aggregation modules, and attention fusion modules. DDRNet [58] enhances information flow between its dual branches by introducing bilateral connections. PIDNet [59] integrates CNN with a proportional-integral-derivative (PID) controller concept, forming a tri-branch network architecture. LWCDNet [35] designs a lightweight dual-branch block in the backbone to extract spatial and contextual information concurrently for cloud detection.

III. Methods

In this section, we first introduce the detailed structure of our TUB illustrated in Fig. 3 and its main components, including the downsampling module (DSM), the SFFM, and the upsampling module (USM). Then, we describe the DSM with TUB (TDSM), which is used to form the encoder branch of our network, and the upsampling fusion module with bilinear (BUSFM), which is used to construct our decoder. Finally, the whole network architecture will be given.

A. Design of TUB

As the conventional networks [18], [37] tend to gradually recover from the semantic representations obtained from the encoder branch by gradually decreasing the spatial size, which seems to work separately, the encoder is not able to do some preparations explicitly for the decode stage. To address this problem, we designed TUB, a lightweight-designed U-shape encoder-decoder structure which captures enriched spatial details from a feature map with specific spatial resolution. Its U-shape structure also guides the encoder to filter and embed extra clues for better recovery during the decoder stage, thus embedding the capability of spatial resolution recovery along the whole branch of the encoder, which helps the decoder to be more efficient and accurate during the recovering phase of our network. As shown in Fig. 3, TUB consists of three parts: DSM, SFFM and USM.

1) DSM: Inside the TUB, n stages of DSM are stacked, which is used as its encoder. For efficient design, DSM is simply constructed by one convention block (ConvBlock) for feature extraction, followed by one Maxpool layer with kernel size and stride both set to 2 for downsampling. As for the ConvBlock shown in the yellow dashed box in Fig. 3, it is a convolution layer with kernel size $k = 3$, stride $s = 1$, and padding $p = 1$, followed by a Group Normalization layer and a ReLU activation function. Given a feature map $F_{\text{in}} \in R^{c\times h\times w}$ as the input of TUB, the output feature map F_{DSM_i} from the ith stage of DSM will, therefore, be

$$F_{\text{DSM}_i} = R^{c'\times \frac{h}{2^i}\times \frac{w}{2^i}} \tag{1}$$

where h, w, and c denote the height, width, and channel size of the input feature map of the TUB, respectively, c' denotes the interchannel size of the DSM, and $i \in \{1, 2, \ldots, n\}$. As shown in Table V, we also demonstrated that $n = 2$ and $c' = 16$ is enough for the cloud detection task; moreover, we keep c' the same in different stages of DSM instead of gradually increasing the channel volume for efficient design.

2) SFFM: As discussed in the recent well-designed model [56], large receptive fields are crucial for segmentation tasks. Thus, as illustrated in the red dashed box in Fig. 3, SFFM is proposed to take both the large receptive field and efficient computation into consideration. For lightweight design, we first apply a single 3×3 dilated convolution with dilation rate and padding, both set as 2 to the feature map F_{DSM_n} to obtain a feature map F'_{DSM_n} with a two times larger receptive field compared to the F_{DSM_n}. Subsequently, a Group Normalization layer and a ReLU activation function will be applied before simply concatenating F_{DSM_n} and F'_{DSM_n} for feature fusion. Finally, we generate the output feature map by passing the fused feature through a ConvBlock; thus, the output feature map $F_{\text{SFFM}} \in R^{2c\times(h/2^n)\times(w/2^n)}$ from the SFFM can be denoted as

$$\begin{aligned} F_{\text{SFFM}} = {} & \text{ConvBlock} \\ & \times \left(\text{concat}\left(F_{\text{DSM}_n}, \text{ReLU}\left(\text{GN}\left(F'_{\text{DSM}_n}\right)\right)\right)\right) \end{aligned} \tag{2}$$

where ReLU and GN denote the ReLU and Group Normalization operations, respectively.

3) USM: The first USM takes F_{SFFM} as input, then the later USM takes the feature map output from the previous USM as input. For lightweight design, USM directly applies a bilinear upsampling layer followed by a ConvBlock used for

smoothening the feature map; thus, the output feature map of the i_{th} USM can be denoted as

$$F_{\text{USM}_i} = \text{ConvBlock}\big(\text{add}\big(F_{\text{DSM}_{n-i+1}}, \text{UP}\big(F_{\text{USM}_{i-1}}\big)\big)\big) \quad (3)$$

where $i \in \{1, 2, \ldots, n\}$, $F_{\text{USM}_i} \in R^{c\times(h/2^{n-i})\times(w/2^{n-i})}$ and add denotes the element-wise summation operation; moreover, it is worth noticing that the number of USM we used will be kept the same as that of DSM since we force TUB to recover the spatial resolution to the same as the input size to capture clues that will benefit later recovery operations along the decoder branch of our network. During the recovery phase, we also use a simple summation operation to introduce high-level information from the encoding phase to achieve high-level guidance for better performance, although it slightly increases the memory access cost in the inference phase. Detailed settings of TUB are shown in Table I.

TABLE I
DETAILED SETTINGS OF TUB

TUB	Stage	Opr	c_{in}	c_{out}	Input size	Output size
Encoder	DSM_1	ConvBlock	64	16	320	320
		Max Pool	-	-	320	160
	DSM_2	ConvBlock	16	16	160	160
		Max Pool	-	-	160	80
SFFM	SFFM	DilatedConv	16	16	80	80
		Concat	-	-	-	-
		ConvBlock	32	16	80	80
Decoder	USM_1	Bilinear up	-	-	80	160
		ConvBlock	64	64	160	160
	USM_2	Bilinear up	-	-	160	320
		ConvBlock	64	64	320	320

B. Design of ANNet

As shown in Fig. 2, our network is mainly composed of four components: a simple input layer, a NE [the blue area in Fig. 2(a)] with TUBs, an ALD [the green area in Fig. 2(a)], and a segmentation head. The NE branch extracts semantic features, which the ALD branch recovers by fusing the up-sampled low-level features from the encoder branch with the high-level features from the counterpart of the decoder branch to lead to better segmentation performance.

1) The Input Layer: It takes the RGB image $I \in R^{3\times H\times W}$ as input, where H and W denote the height and width, respectively, of the input image. We directly use a single ConvBlock to expand the channel volume to C and extract detailed contexts from the input image. Thus, the input layer maps the input image to the feature map $F_{\text{in}} \in R^{C\times H\times W}$. As given in Table V, we demonstrate that $C = 64$ is a good choice for cloud detection tasks.

2) NE Branch: The U-shaped ANNet, in conjunction with the U-shaped TUB, collectively embodies the nested architecture characteristic. First, as illustrated in Fig. 2(b), we construct the TDSM by using a TUB followed by a Maxpool layer. Then, we stack m TDSM to gradually reduce the spatial resolution to encode context information. As shown in Fig. 2, the encoder branch uses TDSM to generate a feature map with downsampling ratios of (1/2), (1/4), and (1/8), respectively. The first TDSM takes F_{in} as input, and the output feature map of the i_{th} stage can be given as

$$F_{\text{enc}_i} \in R^{C\times\frac{H}{2^i}\times\frac{W}{2^i}} \quad (4)$$

where $i \in \{1, 2, \ldots, m\}$. For the last stage, we directly apply TUB to further refine the feature from the spatial resolution at $(H/8) \times (W/8)$, avoiding extra downsampling operation and retaining enough detail spatial information that benefits for recovering operation along the decoder branch; thus, the dimension of the output feature map of the encoder branch will be kept the same as that of F_{enc_m}.

3) Asymmetric and Lightweight Decoder Branch: The main purpose of the decoder branch is to recover spatial resolution from the bottom of the encoder branch for final prediction. As mentioned in [53] and [57], high-level feature guidance and feature fusing are efficient designs in the field of semantic segmentation. BUSFM is, therefore, proposed to achieve a better balance between feature recovery and network lightness. Our decoder branch is formed by the stack p stages of BUSFM. In detail, for the i_{th} stage of BUSFM, the input feature map from the lower layer can be denoted as $F_{\text{low}} \in R^{C\times(H/(m-i+1))\times(W/(m-i+1))}$ and the feature map from the higher layer of the encoder branch can be denoted as $F_{\text{high}} \in R^{C\times(H/(m-i))\times(W/(m-i))}$, where $i \in \{1, 2, \ldots, p\}$. As illustrated in Fig. 2(c), BUSFM first passes F_{low} through a 2×2 transpose convolution with stride $= 2$ to map F_{low} to $F'_{\text{low}} \in R^{(C/2)\times(H/(m-i))\times(W/(m-i))}$, and it applies a single ConvBlock to map F_{high} to $F'_{\text{high}} \in R^{(C/2)\times(H/(m-i))\times(W/(m-i))}$ simultaneously. Then, BUSFM uses a simple concatenation operation for fast aggregating the features with contexts from different spatial ratios. Finally, a ConvBlock will be applied to further refine the fused feature before output. Formally, the output feature map of the ith stage of BUSFM $F_{\text{BUSFM}_i} \in R^{C\times(H/(m-i))\times(W/(m-i))}$ can be derived as

$$F_{\text{BUSFM}_i} = \text{ConvBlock}\big(\text{concat}\big(\text{tconv}(F_{\text{low}}), \text{ConvBlock}\big(F_{\text{high}}\big)\big)\big) \quad (5)$$

where tconv denotes the transpose convolution operation. As given in Table VII, we demonstrate that $p = m - 1$ is an efficient choice for cloud detection tasks.

4) Segmentation Head: It is designed to expand the spatial size of the feature map F_{BUSFM_p} from the top of the decoder branch to the input size $H \times W$, and to adjust the output channel to the desired size C_{out} for final prediction. Thus, in pursuit of efficient design, it directly uses bilinear upsampling followed by a ConvBlock for less computation. Formally, the output feature map of the segmentation head $F_{\text{out}} \in R^{C_{\text{out}}\times H\times W}$ can be derived as

$$F_{\text{out}} = \text{ConvBlock}\big(\text{UP}\big(F_{\text{BUSFM}_p}\big)\big). \quad (6)$$

C. Detailed ANNet Settings

The whole architecture of our proposed network is depicted in Fig. 2, and the detailed network settings are given in Table II.

For the TUB, which is used to construct the TDSM module, $n = 2$ stages of DSM are stacked to embed semantic features,

TABLE II
DETAILED SETTINGS OF ANNET-M

	Stage	Opr	c_{in}	c'	c_{out}	Input size	Output size
Encoder	Input layer	ConvBlock	3	-	64	320	320
	TDSM_1	TUB	64	16	64	320	320
		Max Pool	-	-	-	320	160
	TDSM_2	TUB	64	16	64	160	160
		Max Pool	-	-	-	160	80
	TDSM_3	TUB	64	16	64	80	80
		Max Pool	-	-	-	80	40
	TUB	TUB	64	16	64	40	40
Decoder	BUSFM_1	Bilinear up	-	-	-	80	80
		ConvBlock1	64	-	32	80	80
		ConvBlock2	64	-	32	80	80
		Concat	-	-	64	80	80
	BUSFM_2	Bilinear up	-	-	-	80	160
		ConvBlock1	64	-	32	160	160
		ConvBlock2	64	-	32	160	160
		Concat	-	-	64	160	160
	SegHead	Bilinear up	-	-	-	160	320
		ConvBlock	64	-	2	320	320

followed by one SFFM to fuse feature with multiple receptive fields, and $n = 2$ stages of USM are stacked for feature recovery.

The encoder branch of our network is formed by $m = 3$ stacked TDSM, which shrinks the spatial resolution of its output feature maps with downsampling $(1/2)$, $(1/4)$, and $(1/8)$, then an extra TUB module is applied to strengthen the feature representation before decoding procedures. Subsequently, the decoder branch applies $p = 2$ BUSFM modules to gradually recover the spatial size of the feature maps. Finally, the segmentation head doubled the spatial size of the output feature map from the decoder branch for final prediction.

We, moreover, adopt residual connections between TDSM along the encoder branch to mitigate the gradient vanishing issue in deep neural networks, while $C = 64$ as the input and output channel size will be kept the same in different TDSM modules to give more channel volumes to help combine the information from the residual connection more efficiently. Also, the constant C and the fixed interchannel size of TUB c' together allow our network scale to be flexibly adjusted.

In addition, to further improve the performance of our network, as illustrated in Fig. 2, an extra two segmentation heads are applied to the p feature maps $F_{\text{BUSFM}_{i=1,2,\ldots,p}}$ from the decoder branch to generate p feature map $F'_{\text{BUSFM}_{i=1,2,\ldots,p}}$ for auxiliary training, which only increase minor computation during the training stage while having no effects on the inference phase, and our network will be optimized with cross-entropy loss.

IV. EXPERIMENTS

In this section, we first conduct ablation experiments to demonstrate the efficiency and effectiveness of our proposed ANNet the on GF1_WHU dataset. Subsequently, comparison experiments between our proposed models and the existing state-of-the-art methods are performed on both the validation set of GF1_WHU dataset and 95-Cloud dataset for further performance evaluation.

A. Datasets

1) GF1_WHU Dataset: GF1_WHU (GF-1 Cloud and Cloud Shadow Cover Validation Dataset) [60] is released by the SENDIMAGE Laboratory of Wuhan University and is widely applied in cloud detection, as well as cloud and cloud shadow segmentation tasks. This dataset includes 108 GF-1 wide field of view (WFV) level-2 A scenes of different land cover types and regions collected from May 2013 to August 2016, and the corresponding ground-truth masks fall into four classes: 0 for the background pixel, 1 for the pixel on the clear area, 128 for the pixels on the cloud shadow area and 255 for pixels on the cloud area. Each image in the dataset is composed of four multispectral bands, including three visible R, G, B bands and a near-infrared band, and the spatial resolution of the image is 16 m. In this article, we first extract nonoverlapped 103 906 RGB patches with a size of 512 × 512 by applying a sliding window to the original high-resolution and ground-truth masks. Then, we split the dataset into two sets: 93 516 patches for the training set and 10 390 patches for the validation set. The validation set is also used for the final performance evaluation.

2) 95-Cloud Dataset: 95-Cloud Dataset [61] is a superset of the 38-Cloud Dataset. 38-Cloud Dataset is composed of 38 Landsat eight scene images consisting of four spectral bands: Red (band 4), Green (band 3), Blue (band 2), and Near Infrared (band 5), and its corresponding binary ground-truth masks using pixel values 0 and 1 stand for the background area and cloud areas, respectively. Eight-scene high-resolution images and masks are cropped into patches with a size of 384 × 384, 38-Cloud Dataset is thus comprised of a training set with 8400 patches and a testing set with 9201 patches. The main purpose of this extension is to introduce 57 more scenes from varying types of underlying surfaces and cloud categories to the 38-Cloud Dataset for training to enhance the generalization capability of the models. To the train set, 9201 additional nonoverlapped 384 × 384 training patches with four bands are extracted from 75 Landsat 8 Collection 1 Level-1 scenes mostly located in North America. The test set remains unchanged as in the 38-Cloud Dataset. Thus, the total number of training patches is 34 701, while that in the test set is 9201. Besides, after removing patches that contain all background pixels, the number of valid training patches reduces to 21 502, which ensures that the average cloud coverage of the 95-Cloud dataset remains around 50%. In this article, we separate a validation set with 4301 patches from the whole train set for validation during training and performance evaluation, while the other 17 201 patches are for training.

B. Training and Inference Settings

1) Training Settings: For the sake of fairness, all experiments employ the same training settings, all models are trained from scratch, and the data augmentation strategies for training on GF1_WHU and 95-Cloud datasets will maintain consistency.

In detail, all models using 320 × 320 as input resolution are trained for 40 K iterations on single RTX 2080Ti with batch

size set to 12. The initial learning rate is $1e-4$ of which is adjusted by the cosine annealing scheduler after each iteration. We employ AMSGrad with weight decay, β_1 and β_2 of 0.01, 0.9, and 0.999, respectively, to optimize all models.

As to the data preprocessing, input images are randomly scaled by the ratio range from 0.8 to 1.2, followed by randomly cropping back to 320 × 320 for GF1_WHU dataset, while directly resizing from 384 to 320 used for 95-Cloud dataset. Then, mean and std are applied to the input images before passing into networks.

For the additional data augmentation, 50% of input images will be rotated with a random angle ranged $[-45°, 45°]$, and random flipping with 50% probability is also employed on both GF1_WHU and 95-Cloud dataset.

2) Inference Settings: For fair comparison, the input resolution of all models is fixed to 320 × 320, and inference results, including predicted masks, evaluation metrics and inference speeds of all models are obtained by directly using the Pytorch framework with a single RTX 2080Ti on FP32 without extra optimizing strategies; the batch size is set to 1 for all inferences.

C. Evaluation Metrics

In this article, we assert the performance of existing models and our proposed ANNet by using mAcc, mIoU, and mDice, which are defined as

$$\text{mAcc} = \frac{1}{K+1}\sum_{i=0}^{K}\frac{\text{TP}}{\text{TP}+\text{FP}} \tag{7}$$

$$\text{mDice} = \frac{1}{K+1}\sum_{i=0}^{K}\frac{2\times\text{TP}}{2\times\text{TP}+\text{FP}+\text{FN}} \tag{8}$$

$$\text{mIoU} = \frac{1}{K+1}\sum_{i=0}^{K}\frac{\text{TP}}{\text{FN}+\text{FP}+\text{TP}} \tag{9}$$

where TP, TN, FP, and FN indicate true-positive, true-negative, false-positive, and false-negative predictions.

Additionally, the amounts of model parameters are used to evaluate the complexity of models while GFLOPs for their assessment of computational consumption. FPS is used to evaluate the inference speed of models.

D. Ablative Experiments

In this section, comprehensive experiments are conducted to demonstrate the effective design of our proposed ANNet.

1) Effective Design of TUB: We first demonstrate the effectiveness of the U-shape design of TUB. Recent proposed methods like [53], [54] proved that dual-branch structures incline to be sorts of effective lightweight methods in the field of semantic segmentation; thus, we adapt our TUB to two kinds of dual-branch structures illustrated in Fig. 4(a) and (b). The results in Table III show us that TUB slightly outperforms that using a dual-branch structure, and it also has a slight advantage in model size and computational efficiency.

Besides, we replace the decoder branch of TUB with direct upsampling depicted in Fig. 3 to verify the necessity of the U-shape architecture of TUB. According to the results in Table III, TUB with direct upsampling (TUB-DU) cuts the performance to 93.16 (−0.15%) compared to that with a U-Shape structure, which proves that TUB with a U-shape architecture is crucial for ANNet.

2) Scale of TUB: We then explore the appropriate number of stages n inside the TUB, as varying n results in diverse scales that affect the capability of enriching semantic context from each stage with a different spatial ratio of the encoder. According to the results given in Table IV, a rapid increase in parameter volume of ANNet cannot be observed by varying n from 2 to 4 due to the efficient design of TUB discussed in Section III-A. The performances of mAcc, mDice, and mIoU, nevertheless, remain steady. For the balance between computation and performance, we, therefore, choose $n = 2$ for our ANNet.

TABLE III

ABLATION STUDY OF THE EFFECTIVE DESIGN OF TUB ON ANNET

structure	Params(M)	GFLOPs	mAcc	mDice	mIoU
TUB	0.240	**5.007**	**96.65**	**96.50**	**93.31**
TUB-DU	**0.231**	3.979	96.52	96.42	93.16
DB-1	0.240	5.061	96.58	96.46	93.24
DB-2	0.287	5.285	96.59	96.48	93.26

DB-1 and DB-2 denote the TUB with adapted dual-branch structure illustrated in Fig. 4. TUB-DU denotes the adapted TUB illustrated in Fig. 3 that uses direct up-sampling instead of a U-shape structure.

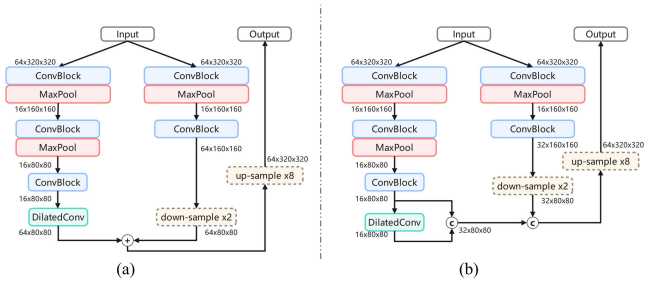


Fig. 4. Two different designs of dual-branch structure. (a) Design of DB-1. (b) Design of DB-2.

TABLE IV

ABLATION STUDY OF THE SCALE OF TUB ON ANNET

n	Params(M)	GFLOPs	mAcc	mDice	mIoU
2	**0.240**	5.007	96.65	**96.50**	**93.31**
3	0.259	5.003	96.58	96.49	93.28
4	0.278	**5.002**	**96.66**	96.50	93.30

n denotes the number of the stages of the encoder and decoder of TUB.

TABLE V

ABLATION STUDY OF THE CHANNEL CAPACITY OF TUB ON ANNET

c_{in}	c'	c_{out}	Params	GFLOPs	mAcc	mDice	mIoU
16	8	16	**29.506k**	**0.601**	95.15	95.07	90.74
32	8	32	61.346k	1.364	95.69	95.51	91.52
32	16	32	0.115M	2.167	96.09	95.91	92.23
64	16	64	0.240M	5.007	96.65	96.50	93.31
64	32	64	0.453M	8.188	96.77	96.65	93.57
128	32	128	0.949M	19.128	**96.81**	**96.74**	**93.74**
deeper	*deeper*	*deeper*	3.226M	21.456	96.42	96.37	93.06

c_{in} and c_{out} are the input and output channel volumes of TUB, and c' is the interchannel volume of TUB. *deeper* denotes c_{in}, c', and c_{out} of TUB in i_{th} stage of encoder is set to 2^{4+i}, 2^{5+i}, and 2^{3+i} respectively, where $i \in \{1, 2, 3, 4\}$.

It is worth to notice that the computational cost of ANNet slightly decrease with the growth of n. That is due to the increment of computation introduced by the additional DSM and USM being less than the decrement of the computation amount of SFFM. Specifically, as shown in Fig. 3, one extra DSM and USM will be added above SFFM whenever n increases by 1, which results in extra computation cost $\Delta_a = 2 \cdot f_w f_h \cdot (k^2 c'^2 + 5c')$ while SFFM requires $\Delta_b = (1 - 2^{-2}) \cdot (f_w f_h \cdot (3k^2 c'^2 + 10c'))$ fewer computation as the resolution of its input reduces by half. We then have $\Delta = \Delta_a - \Delta_b = f_w f_h \cdot (-(1/4)k^2 c'^2 + (10/4)c') < 0$, which shows a slight decrease of the computation of TUB although its parameter amount rises by $2 \cdot (k^2 c'^2 + 3c')$. Besides, f_w and f_h here denote the width and height of the input feature map of the inserted DSM/USM, respectively, $k = 3$ and $c' = 16$ denote the kernel size and the inter-channel size, respectively, as discussed in session III. This also demonstrates the effectiveness of the design of our TUB and ANNet.

3) Channel Capacity of TUB: As discussed in Section III-A, stacked TUB with constant c_{in}, c' and c_{out} is used as the encoder of ANNet, which allows the ability of spatial detail extraction of ANNet to be scaled flexibly. Varying c_{in}, c' or c_{out} affects the total channel capacity of TUB, so as to the representation ability of the encoder of ANNet; hence, we mainly scale ANNet from the aspects of c_{in}, c_{out}, and c' to investigate the appropriate channel capacity of ANNet for cloud detection tasks. As results are given in the first six rows of Table V, a step increase in c_{in}, c' or c_{out} results in a linear upward trend of model size and GFLOPs while the growth of mIoU slows down gradually; thus, for the balancing between computational cost and performance, we choose $c_{\text{in}} = c_{\text{out}} = 64, c' = 16$ for our ANNet and term it ANNet-m. Besides, we choose $c_{\text{in}} = c_{\text{out}} = 16, c' = 8$ to construct a smaller model named as ANNet-s in the pursuit of extremely lightweight design and high-speed inference while generalizing ANNet to a larger model named as ANNet-l for better performance by choosing $c_{\text{in}} = c_{\text{out}} = 128, c' = 32$.

What is more, as discussed in Section III-B, vary from the modern proposed methods tend to reduce spatial resolution while gradually expanding the channel volume of the feature map as the network goes deeper to capture semantic contexts. We keep channel capacity unchanged for each stage of the encoder in ANNet for further lightweight design. To verify the effectiveness of this approach, we conduct an extra experiment using $c_{\text{in}} = [32, 64, 128, 256], c' = [16, 32, 64, 128]$, and $c_{\text{out}} = [64, 128, 256, 256]$ to construct ANNet-deeper that gradually increase channel size while the network goes deeper. As shown in the last row of Table V, a 0.25% decrease in mIoU can be observed, although over 13× and four times surges in the parameter amount and GFLOPs of ANNet-deeper compared to our ANNet-m, respectively, which also proves that constant $c_{\text{in}} = c_{\text{out}} = 64, c' = 16$ are already sufficient for cloud detection tasks.

4) Effective Design of SFFM Inside the TUB: As discussed in Section III-B, the proposed lightweight SFFM is employed at the bottom of the encoder branch of TUB to expand the receptive field and fuse features with different receptive fields for further feature refining. In order to verify the effectiveness of SFFM, we first perform experiments using different dilation rates and different numbers of dilation convolution layers to generate feature maps with diverse spatial scales. Second, we also adapt aspp borrowed from deeplab v2 [16] to our SFFM to extract semantic features from multiple spatial sizes by using dilated convolution more effectively. Finally, we also investigate different simple methods to fuse the feature maps before and after SFFM by replacing the concatenation operation with an element-wise addition operation.

TABLE VI

ABLATION STUDY OF THE EFFECTIVE DESIGN OF SFFM INSIDE THE TUB ON ANNET-M

dr	*aspp*	*add*	*cat*	Params(M)	GFLOPs	mAcc	mDice	mIoU
				0.212	**4.947**	96.58	96.44	93.19
[2]		√		0.231	4.987	96.61	96.46	93.22
[2]			√	0.240	5.007	96.65	96.50	93.31
[2,4]			√	0.259	5.047	96.64	96.46	93.23
[4]			√	0.240	5.007	96.58	96.43	93.26
	√		√	0.248	5.024	**96.65**	**96.52**	**93.34**
	√			0.230	4.984	96.63	96.50	93.29

dr denotes the dilation rate of each dilated convolution layer used by BCB inside SFFM. *aspp* denotes using aspp instead of dilation convolution to expand the receptive field of the feature map in SFFM. *cat* denotes the feature fusing operation to concatenate the input feature map and the output feature map of SFFM. a*dd* denotes the feature fusing operation to perform element-wise addition to the input feature map and the output feature map of SFFM.

TABLE VII

ABLATION STUDY OF THE DEPTH OF ENCODER AN DECODER ON ANNET-M

m	*p*	*r*	Params(M)	GFLOPs	mAcc	mDice	mIoU
3	2	×1	0.210	10.523	96.63	96.48	93.26
4	1	×4	**0.183**	3.520	96.55	96.41	93.13
4	2	×2	0.240	5.007	96.65	96.50	93.31
5	2	×4	0.270	3.621	96.62	96.49	93.29
6	2	×8	0.301	**3.275**	96.47	96.36	93.04
4	3	×1	0.298	10.928	**96.66**	**96.56**	**93.41**
5	3	×2	0.328	5.108	96.62	96.52	93.34
5	4	×1	0.386	11.029	96.64	96.54	93.38

m denotes the number of encoder layers of ANNet-m. *p* denotes the number of decoder layers of ANNet-m. *r* denotes the up-sample ratio for the feature map from the last decoder to that of the final output feature map of ANNet-m.

According to the results shown in Table VI, both using one dialed convolution with dilation rate dr = 4 and using two consecutive dialed convolutions with dr = 2, 4 tend to impede the computational efficiency and performance of ANNet-m. Although using aspp brings a slight improvement to mIoU, it introduces additional parameters and increases memory access costs, which hamper the efficiency of our network. Besides the fusing operation, concatenation outperforms addition by almost 0.1% on mIoU; thus, our SFFM adopts one dilated convolution layer with dr = 2 and the concatenation operation for feature fusing.

5) Depth of Encoder: A deeper encoder is likely to be more powerful for semantic feature extraction; nevertheless, its performance is possible to be hindered by insufficient training; thus, to investigate the suitable depth of the encoder of ANNet-m, we perform experiments by varying m stages of encoder layers in ANNet-m from 3 to 6 with fixed $p = 2$. As the results given in the first four rows of Table VII, mIoU grows initially, followed by a decrease, and peaks at

TABLE VIII
ABLATION STUDY OF THE IMPACTS OF USING DIFFERENT UPAMPLING METHODS WITH ANNET-M

bilinear	*bilinear-UAFM*	*tp-conv*	*sp-conv*	Params (M)	GFLOPs	mAcc	mDice	mIoU
✓				0.240	5.007	**96.65**	**96.50**	**93.31**
	✓			**0.211**	**4.420**	96.52	96.42	93.15
		✓		0.252	4.999	96.57	96.44	93.19
			✓	0.663	7.084	96.45	96.36	93.04

bilinear, *tp-conv* and *sp-conv* denote that using bilinear-upsampling, transpose convolution and sub-pixel convolution for spatial resolution recovery, respectively. *bilinear-UAFM* denotes that using bilinear-upsampling with UFAM.

TABLE IX
ABLATION STUDY OF THE IMPACTS OF USING EXTRA LOSS FOR TRAINING ANNET-M

CE	DICE	OHEM	mAcc	mDice	mIoU
✓			96.65	96.50	93.31
✓	✓		96.61	96.46	93.23
✓		✓	**96.77**	**96.54**	**93.38**
✓	✓	✓	96.77	96.50	93.30

CE, *DICE* and *OHEM* denote Cross-Entropy loss, Dice loss and online hard example mining (OHEM) respectively.

93.31 when $m = 4$. Similarly, a decrease trend can also be observed when we increase m from 4 to 5 with $p = 3$; hence, our experiments prove that $m = 4$ and $p = 2$ are the best selections of our ANNet-m.

6) Effectiveness of the ALD: A certain number of recent works employed the same number of decoder layers that the encoder used to recover spatial resolution gradually; however, the decoder layers in lower levels occupy a heavier number of parameters, which means that more decoder layers in lower levels tend to make the network harder to train. To pursue lightweight design, we carry out experiments to discover the optimal depth of ALD by using different $p = 1, 2, 3$ and $p = 2, 3, 4$ with $m = 4$ and $m = 5$, respectively. Although the results in Table VII confirm the effectiveness of the decoding process, they also show that GFLOPs rise rapidly with the growth of p; therefore, we choose $p = 2$ to strike the balance between efficiency and performance for our ANNet-m.

7) Impacts of Upsampling Methods: Recent works [62], [63] demonstrated to us that transpose convolution and subpixel convolution are effective methods for spatial context recovery during decoding stages. Compared to bilinear upsampling, the major advantage of transpose convolution and subpixel convolution is that they can be trained with the whole network; nevertheless, they both introduce extra parameters and computations. To investigate the impacts of upsampling methods on our ANNet-m, we perform experiments using these three approaches for our decoder. Besides, following PP-LiteSeg [57], we adopt the lightweight UAFM with our proposed BUSFM for fusing the high-level and low-level semantic features for better recovery guidance. From the results in Table VIII, the bilinear upsampling with no extra trainable parameters even outperforms the transpose convolution and subpixel convolution by 0.12% and 0.27% on mIoU, respectively, while a 0.13% decline in mIoU is observed after involving UAFM. Thus, we directly employ bilinear upsampling without UAFM for recovery during the decoding stages.

8) Impacts of Extra Losses: Finally, we attempt to employ dice loss as an extra loss and introduce online hard example mining (OHEM) in the training stage to further optimize ANNet-m. According to the results in Table IX, dice loss seems to not provide additional benefit while OHEM brings slight improvements to mAcc, mDice, and mIoU as well. As a result, cross-entropy loss with OHEM is adopted.

E. Comparison With the State-of-the-Art Methods

In this section, we conduct experiments for comparisons between our proposed model and the existing state-of-the-art models on GF1_WHU and 95-Cloud datasets. As shown in Table X, our experiments are separated into four groups: nonreal-time models, lightweight real-time models, cloud detection models, and our proposed models. Ours include ANNet-s, ANNet-m and ANNet-l and that trained with OHEM marked with *. All metrics, including mAcc, mDice, mIoU, and the inference speed, are measured on a single RTX 2080Ti, with batch size set to 1 and the input size set to 320 × 320 for all models for fair comparison.

From the results presented in Table X, our ANNet-m* achieves 96.77%, 96.54% and 93.38% on mAcc, mDice and mIoU, respectively, with 221 FPS, while our ANNet-l* even yield the numbers at 96.96%, 96.77%, and 93.79% with an inference speed still up to 125 FPS, which is the state-of-the-art on GF1_WHU dataset. Likewise, similar results can be revealed on 95-Cloud dataset; the performance of ANNet-l* on mAcc, mDice, and mIoU also reach 94.98%, 94.92%, and 90.34, respectively; moreover, the mIoU of our extremely lightweight ANNet-s* with only 29.506 K parameters can remain at the level of 90.74% on GF1_WHU and 84.38% on 95-Cloud datasets, which further proves the effectiveness of our proposed ANNet; urthermore, for the purpose of highlighting the effectiveness of our ANNet, some detection results of GF1_WHU and 95-Cloud datasets are visualized in Figs. 5–8. Among them, Figs. 5 and 6 illustrate the results of real-time semantic segmentation methods and cloud detection methods on GF1_WHU dataset; Figs. 7 and 8 are the results of their study on 95-Cloud dataset.

We select four challenging areas for cloud detection, namely green land (Row 1), barren land (Row 3), water surface (Row 5), and scenes where clouds coexist with snow (Row 7). To better highlight the details, we crop out the key parts (indicated by red boxes) from the odd-numbered rows and display them alongside in the corresponding even-numbered rows.

In Figs. 5 and 7, methods from (f) to (j) not only fail to handle the details well at the edge of the cloud but also miss small regions. Methods Figs. 5 and 7(h), (i), and (j) have frequent misdetections at the water-land boundary. In Figs. 6 and 8, these cloud detection methods perform better at the edges of clouds compared to real-time semantic segmentation methods, offering richer detail features; however, cloud detection methods also have more misdetections, such as (c). Despite having

TABLE X
RESULTS OF COMPARISION EXPERIMENTS ON THE VALIDATION SET OF GF1_WHU AND 95-CLOUD DATASETS

Year	Method	Params (M)	GFLOPs	FPS	GF1_WHU mAcc	GF1_WHU mDice	GF1_WHU mIoU	95-Cloud mAcc	95-Cloud mDice	95-Cloud mIoU
				Non-real-time models						
2015	UNet [18]	28.987	79.485	-	96.07	96.09	92.55	94.37	94.26	89.14
2017	PSPnet [14]	48.982	71.680	-	96.14	95.78	91.99	94.02	93.80	88.34
2018	Att-UNet [38]	34.879	106.496	-	96.10	96.07	92.52	94.51	94.38	89.36
2019	UNet++ [37]	36.630	221.184	-	94.82	95.27	91.09	90.56	90.71	83.01
2020	U2-Net [36]	44.024	59.015	-	94.98	95.10	90.77	93.28	93.11	87.12
2021	Swin-UNet [43]	41.424	18.886	-	96.06	96.11	92.59	92.96	92.97	86.88
2022	UNeXt [46]	1.472	0.911	-	96.03	96.03	92.45	93.69	93.66	88.08
2023	DA-TransUNet [47]	110.592	63.517	-	96.00	96.41	93.13	93.90	93.58	87.94
				Real-time models						
2016	ENet [48]	0.349	0.751	103	95.11	95.34	91.21	90.27	89.83	81.54
2018	BiSeNet v1 [53]	13.271	5.789	249	95.72	95.79	92.01	91.99	91.73	84.74
2020	BiSeNet v2 [54]	3.341	4.799	188	96.00	95.54	91.56	91.78	91.47	84.28
2021	CABiNet [55]	5.194	1.028	140	94.08	93.94	88.75	90.20	90.13	82.04
2021	STDC1 [56]	8.271	3.298	224	95.34	95.14	90.84	92.77	92.66	86.34
2021	STDC2 [56]	12.301	4.589	161	94.70	94.78	90.22	91.28	91.11	83.67
2022	PP-LiteSeg [57]	11.273	1.685	197	94.03	93.74	88.41	90.95	90.77	83.11
2022	DDRNet [58]	20.295	7.005	181	95.47	95.53	91.55	92.10	92.04	85.27
2023	PIDNet-s [59]	7.716	2.317	143	92.92	93.83	88.57	86.65	86.98	77.02
2023	PIDNet-m [59]	28.759	8.687	142	94.34	94.75	90.17	88.67	88.91	80.06
				Cloud detection models						
2019	CloudSegNet[21]	5.4K	0.304	-	92.22	93.29	87.65	83.60	83.78	72.14
2020	CloudUNet[22]	31.334	25.677	-	95.43	95.69	91.63	91.97	91.98	85.16
2019	CDNet v1[20]	85.040	101.550	-	94.93	95.44	91.39	93.52	93.33	87.49
2020	CDNet v2[23]	26.863	28.230	-	96.00	96.27	92.88	94.64	94.60	89.75
2023	BoundaryNet[26]	89.423	124.090	-	95.19	94.68	90.04	92.57	92.30	85.70
				Ours						
	ANNet-s	29.506K	0.601	251	95.15	95.07	90.74	91.32	91.34	84.07
	ANNet-m	0.240	5.007	221	96.65	96.50	93.31	94.31	94.30	89.22
	ANNet-l	0.949	19.128	125	96.81	96.74	93.74	94.72	94.71	89.95
	ANNet-s*	**29.506K**	**0.601**	**251**	95.68	95.17	90.90	91.53	91.52	84.38
	ANNet-m*	0.240	5.007	221	96.77	96.54	93.38	94.41	94.36	89.32
	ANNet-l*	0.949	19.128	125	**96.96**	**96.77**	**93.79**	**94.98**	**94.92**	**90.34**

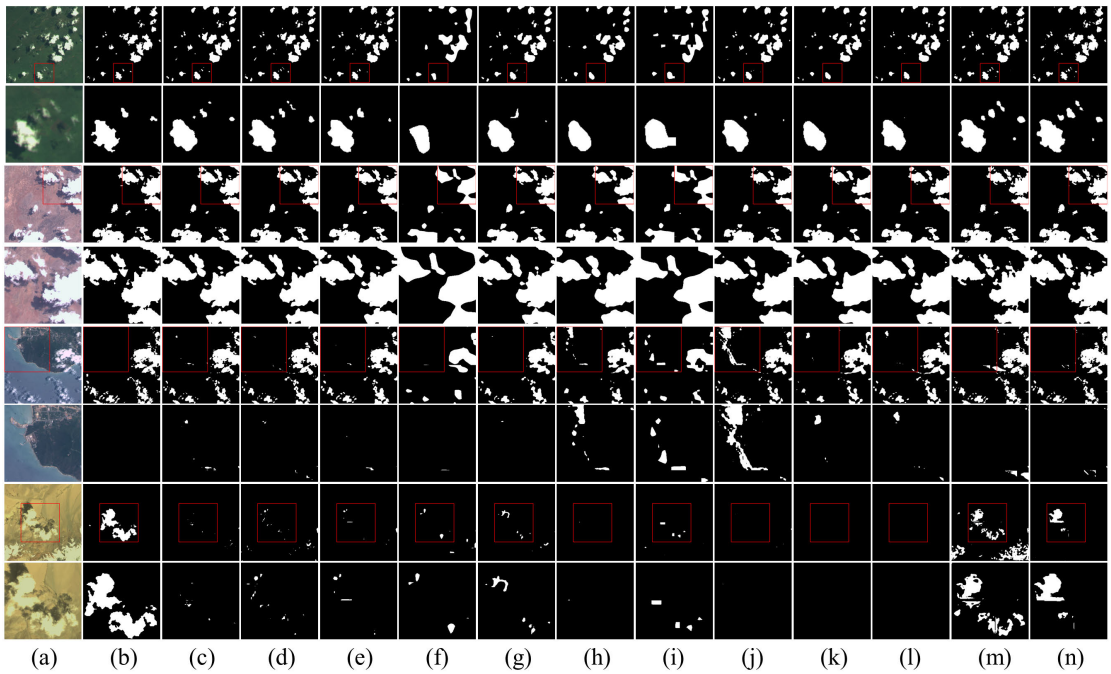


Fig. 5. Results of real-time semantic segmentation methods on GF1_WHU dataset. (a) Input image. (b) Ground truth. (c) ENet. (d) BiSeNet v1. (e) BiSeNet v2. (f) CABiNet. (g) STDC1. (h) STDC2. (i) PP-LiteSeg. (j) DDRNet. (k) PIDNet-s. (l) PIDNet-m. (m) ANNet-s. (n) ANNet-m.

only 5.4 k parameters and 0.304 GFlops, making it the most lightweight among all the methods compared, the visual results of (c) are not satisfactory. This is because it employs SegNet as the backbone for extracting multiscale features, a process that is too simple and direct, leading to a significant loss of detail during the downsampling process. It demonstrates that a

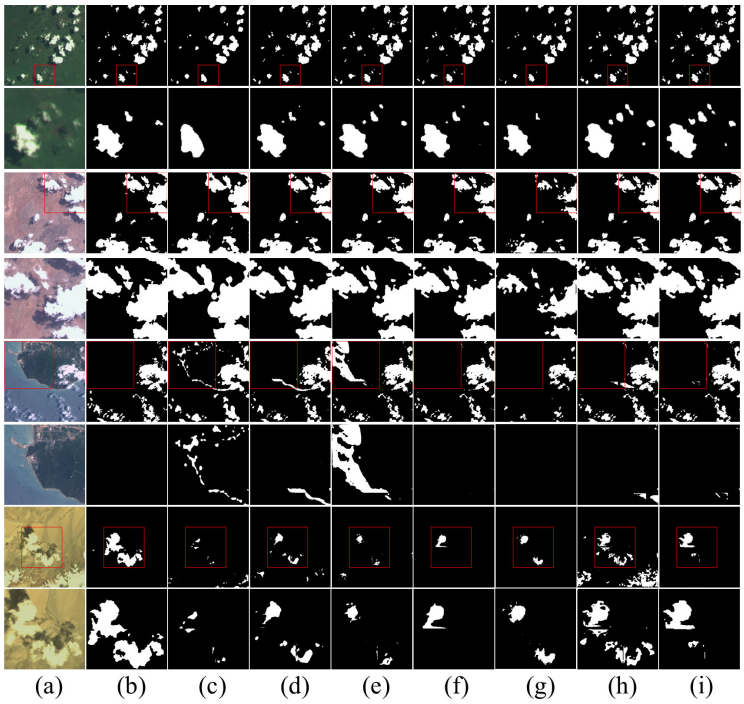


Fig. 6. Results of cloud detection methods on GF1_WHU dataset. (a) Input image. (b) Ground Truth. (c) CloudSegNet. (d) CloudUNet. (e) CDNet v1. (f) CDNet v2. (g) BoundaryNet. (h) ANNet-s. (i) ANNet-m.

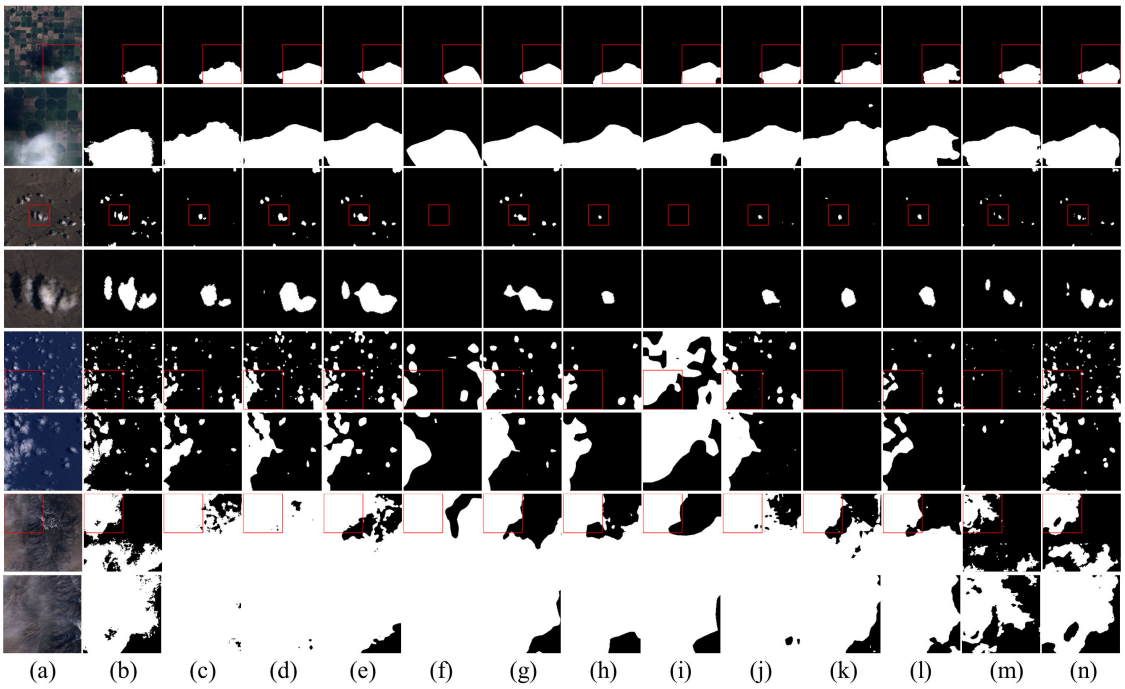


Fig. 7. Results of real-time semantic segmentation methods on 95-Cloud dataset. (a) Input image. (b) Ground truth. (c) ENet. (d) BiSeNet v1. (e) BiSeNet v2. (f) CABiNet. (g) STDC1. (h) STDC2. (i) PP-LiteSeg. (j) DDRNet. (k) PIDNet-s. (l) PIDNet-m. (m) ANNet-s. (n) ANNet-m.

more lightweight model does not necessarily equate to better performance.

Compared to all other methods, the proposed ANNet detects cloud edges with richer details and seldom misses small-sized objects in scenes where fragmented clouds are dense. Notably, ANNet also retains a certain capability to identify clouds under the complex scenario of coexisting clouds and snow, which is seen as the most challenging task in cloud detection, demonstrating good robustness than those cloud detection methods. This highlights that the cloud detection performance of the proposed network surpasses existing methods.

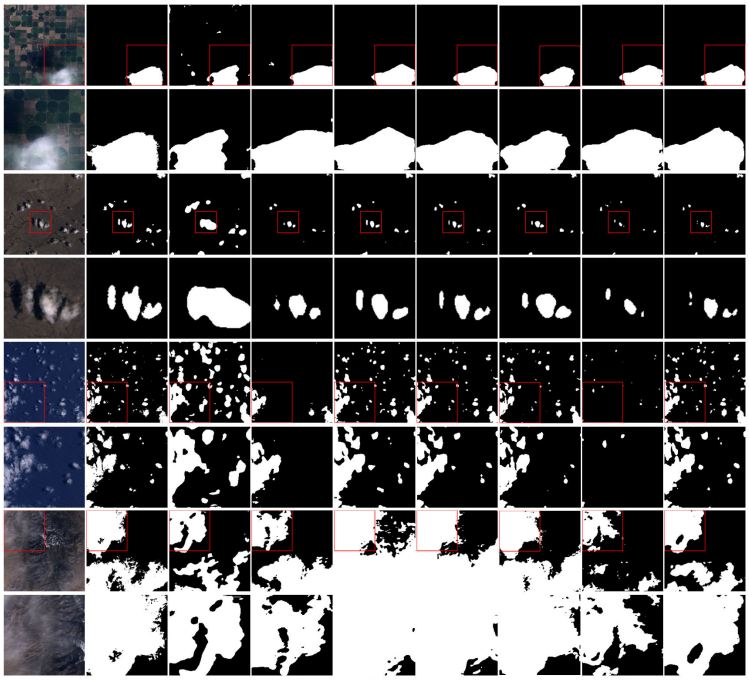


Fig. 8. Results of cloud detection methods on 95-Cloud dataset. (a) Input image. (b) Ground truth. (c) CloudSegNet. (d) CloudUNet. (e) CDNet v1. (f) CDNet v2. (g) BoundaryNet. (h) ANNet-s. (i) ANNet-m.

V. Conclusion

In this article, ANNet is proposed to trade off segmentation accuracy performance against computation cost. TUB is designed to effectively extract spatial details from different encode stages while allowing ANNet to learn recovery abilities earlier in the encode phase compared to most existing approaches. In the decode phase, ALD with simple structures is introduced for effective recovery with high-level context guidance from the counterpart encoder. Extensive experiments are, moreover, conducted to prove that it is not necessary to use larger channel capacity in deeper layers in cloud detection tasks, which benefits shrinking model parameters and achieving faster inference speeds. Finally, by mainly adjusting the channel capacity, we have ANNet-s, ANNet-m, and ANNet-l. For the sake of the sophisticated design of the architecture for cloud detection, excellent performance can be observed in the results of our comparison experiments shown in Table X and the visual segmentation results in from Figs. 5 to 8. ANNet-l achieves state-of-the-art performance with real-time inference speed, while ANNet-s is even able to infer at 251 FPS when mIoU is still approaching 91%. Our further research will focus on extending our method to multiple classes semantic segmentation and other tasks in the field of remote sensing.

References

[1] Y. Zhang, W. B. Rossow, A. A. Lacis, V. Oinas, and M. I. Mishchenko, "Calculation of radiative fluxes from the surface to top of atmosphere based on ISCCP and other global data sets: Refinements of the radiative transfer model and the input data," *J. Geophys. Res., Atmos.*, vol. 109, no. D19, p. 2003, Oct. 2004, doi: 10.1029/2003jd004457.

[2] G. Gesell, "An algorithm for snow and ice detection using AVHRR data an extension to the Apollo software package," *Int. J. Remote Sens.*, vol. 10, nos. 4–5, pp. 897–905, Apr. 1989, doi: 10.1080/01431168908903929.

[3] L. L. Stowe et al., "Global distribution of cloud cover derived from NOAA/AVHRR operational satellite data," *Adv. Space Res.*, vol. 11, no. 3, pp. 51–54, 1991, doi: 10.1016/0273-1177(91)90402-6.

[4] L. Wang, Y. Chen, L. Tang, R. Fan, and Y. Yao, "Object-based convolutional neural networks for cloud and snow detection in high-resolution multispectral imagers," *Water*, vol. 10, no. 11, p. 1666, Nov. 2018, doi: 10.3390/w10111666.

[5] C. Latry, C. Panem, and P. Dejean, "Cloud detection with SVM technique," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Barcelona, Spain, Jan. 2007, pp. 448–451, doi: 10.1109/IGARSS.2007.4422827.

[6] J. Deng, H. Wang, and J. Ma, "An automatic cloud detection algorithm for Landsat remote sensing image," in *Proc. 4th Int. Workshop Earth Observ. Remote Sens. Appl. (EORSA)*, Guangzhou, China, Jul. 2016, pp. 395–399, doi: 10.1109/EORSA.2016.7552837.

[7] L. Xu, A. Wong, and D. A. Clausi, "A novel Bayesian spatial–temporal random field model applied to cloud detection from remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 4913–4924, Sep. 2017, doi: 10.1109/TGRS.2017.2692264.

[8] Q. Li, W. Lu, J. Yang, and J. Z. Wang, "Thin cloud detection of all-sky images using Markov random fields," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 417–421, May 2012, doi: 10.1109/LGRS.2011.2170953.

[9] Z. Guo, C. Li, Z. Wang, E. Kwok, and X. Wei, "A cloud boundary detection scheme combined with ASLIC and CNN using ZY-3, GF-1/2 satellite imagery," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 455–458, Apr. 2018, doi: 10.5194/isprs-archives-xlii-3-455-2018.

[10] L. Zhang et al., "Improving semantic image segmentation with a probabilistic superpixel-based dense conditional random field," *IEEE Access*, vol. 6, pp. 15297–15310, 2018, doi: 10.1109/ACCESS.2018.2814568.

[11] S. Le Hégarat-Mascle and C. André, "Use of Markov random fields for automatic cloud/shadow detection on high resolution optical images," *ISPRS J. Photogramm. Remote Sens.*, vol. 64, no. 4, pp. 351–366, Jul. 2009, doi: 10.1016/j.isprsjprs.2008.12.007.

[12] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017, doi: 10.1109/TPAMI.2016.2572683.

[13] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017, doi: 10.1109/TPAMI.2016.2644615.

[14] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239, doi: 10.1109/CVPR.2017.660.

[15] C. Liang-Chieh, G. Papandreou, I. Kokkinos, K. Murphy, and A. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," in *Proc. Int. Conf. Learn. Represent.*, San Diego, CA, USA, May 2015, pp. 1–12. [Online]. Available: https://inria.hal.science/hal-01263610

[16] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, Atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018, doi: 10.1109/TPAMI.2017.2699184.

[17] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking Atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

[18] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, vol. 9351, Munich, Germany, 2015, pp. 234–241, doi: 10.1007/978-3-319-24574-4_28.

[19] J. Qian, J. Ci, H. Tan, W. Xu, Y. Jiao, and P. Chen, "Cloud detection method based on improved DeeplabV3+ remote sensing image," *IEEE Access*, vol. 12, pp. 9229–9242, 2024, doi: 10.1109/ACCESS.2024.3353205.

[20] J. Yang, J. Guo, H. Yue, Z. Liu, H. Hu, and K. Li, "CDNet: CNN-based cloud detection for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6195–6211, Aug. 2019, doi: 10.1109/TGRS.2019.2904868.

[21] S. Dev, A. Nautiyal, Y. H. Lee, and S. Winkler, "CloudSegNet: A deep network for nychthemeron cloud image segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 12, pp. 1814–1818, Dec. 2019, doi: 10.1109/LGRS.2019.2912140.

[22] C. Shi, Y. Zhou, B. Qiu, D. Guo, and M. Li, "CloudU-Net: A deep convolutional neural network architecture for daytime and nighttime cloud images' segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1688–1692, Oct. 2021, doi: 10.1109/LGRS.2020.3009227.

[23] J. Guo, J. Yang, H. Yue, H. Tan, C. Hou, and K. Li, "CDNetV2: CNN-based cloud detection for remote sensing imagery with cloud-snow coexistence," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 700–713, Jan. 2021, doi: 10.1109/TGRS.2020.2991398.

[24] C. Zhang, L. Weng, L. Ding, M. Xia, and H. Lin, "CRSNet: Cloud and cloud shadow refinement segmentation networks for remote sensing imagery," *Remote Sens.*, vol. 15, no. 6, p. 1664, Mar. 2023, doi: 10.3390/rs15061664.

[25] Q. He, X. Sun, Z. Yan, and K. Fu, "DABNet: Deformable contextual and boundary-weighted network for cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601216, doi: 10.1109/TGRS.2020.3045474.

[26] C. Zhao, X. Zhang, N. Kuang, H. Luo, S. Zhong, and J. Fan, "Boundary-aware bilateral fusion network for cloud detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403014, doi: 10.1109/TGRS.2023.3276750.

[27] K. Wu, Z. Xu, X. Lyu, and P. Ren, "Cloud detection with boundary nets," *ISPRS J. Photogramm. Remote Sens.*, vol. 186, pp. 218–231, Apr. 2022, doi: 10.1016/j.isprsjprs.2022.02.010.

[28] J. Li and Q. Wang, "CSDFormer: A cloud and shadow detection method for Landsat images based on transformer," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103799, doi: 10.1016/j.jag.2024.103799.

[29] B. Zhang, Y. Zhang, Y. Li, Y. Wan, and Y. Yao, "CloudViT: A lightweight vision transformer network for remote sensing cloud detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2022.3233122.

[30] Z. Hu, L. Weng, M. Xia, K. Hu, and H. Lin, "HyCloudX: A multibranch hybrid segmentation network with band fusion for cloud/shadow," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 6762–6778, 2024, doi: 10.1109/JSTARS.2024.3374233.

[31] C. Gong, T. Long, R. Yin, W. Jiao, and G. Wang, "A hybrid algorithm with swin transformer and convolution for cloud detection," *Remote Sens.*, vol. 15, no. 21, p. 5264, Nov. 2023, doi: 10.3390/rs15215264.

[32] J. Li et al., "A lightweight deep learning-based cloud detection method for Sentinel-2A imagery fusing multiscale spectral and spatial features," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5401219, doi: 10.1109/TGRS.2021.3069641.

[33] K. Chen, M. Xia, H. Lin, and M. Qian, "Multiscale attention feature aggregation network for cloud and cloud shadow segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612216, doi: 10.1109/TGRS.2023.3283435.

[34] J. Zhang, J. Wu, H. Wang, Y. Wang, and Y. Li, "Cloud detection method using CNN based on cascaded feature attention and channel attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4104717, doi: 10.1109/TGRS.2021.3120752.

[35] C. Luo et al., "LWCDNet: A lightweight network for efficient cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409816, doi: 10.1109/TGRS.2022.3173661.

[36] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O. R. Zaiane, and M. Jagersand, "U^2-Net: Going deeper with nested U-structure for salient object detection," *Pattern Recognit.*, vol. 106, Oct. 2020, Art. no. 107404, doi: 10.1016/j.patcog.2020.107404.

[37] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-net architecture for medical image segmentation," in *Proc. DLMIA*, Granada, Spain, 2018, pp. 3–11, doi: 10.1007/978-3-030-00889-5_1.

[38] O. Oktay et al., "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*.

[39] H. Chen et al., "Slice-to-slice context transfer and uncertain region calibration network for shadow detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 166–182, Sep. 2023, doi: 10.1016/j.isprsjprs.2023.07.027.

[40] Q. Zhu, Y. Yang, X. Sun, and M. Guo, "CDANet: Contextual detail-aware network for high-spatial-resolution remote-sensing imagery shadow detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617415, doi: 10.1109/TGRS.2022.3143886.

[41] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent.*, 2021, pp. 1–11. [Online]. Available: https://iclr.cc/virtual/2021/papers.html?filter=titles

[42] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 9992–10002, doi: 10.1109/ICCV48922.2021.00986.

[43] H. Cao et al., "Swin-UNet: UNet-like pure transformer for medical image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 205–218.

[44] J. Chen et al., "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.

[45] Y. Zhang, H. Liu, and Q. Hu, "TransFuse: Fusing transformers and CNNs for medical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, 2021, pp. 14–24, doi: 10.1007/978-3-030-87193-2_2.

[46] J. Valanarasu and V. M. Patel, "UNeXt: MLP-based rapid medical image segmentation network," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. (MICCAI)*, Singapore, 2022, pp. 23–33, doi: 10.1007/978-3-031-16443-9_3.

[47] G. Sun et al., "DA-TransUNet: Integrating spatial and channel dual attention with transformer U-net for medical image segmentation," *Frontiers Bioeng. Biotechnol.*, vol. 12, May 2024, Art. no. 1398237, doi: 10.3389/fbioe.2024.1398237.

[48] A. Paszke, A. Chaurasia, S. Kim, and E. Culurciello, "ENet: A deep neural network architecture for real-time semantic segmentation," 2016, *arXiv:1606.02147*.

[49] H. Zhao, X. Qi, X. Shen, J. Shi, and J. Jia, "ICNet for real-time semantic segmentation on high-resolution images," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, Sep. 2018, pp. 418–434, doi: 10.1007/978-3-030-01219-9_25.

[50] M. Oršic, I. Krešo, P. Bevandic, and S. Šegvic, "In defense of pre-trained ImageNet architectures for real-time semantic segmentation of road-driving images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 12599–12608, doi: 10.1109/CVPR.2019.01289.

[51] H. Li, P. Xiong, H. Fan, and J. Sun, "DFANet: Deep feature aggregation for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 9514–9523, doi: 10.1109/CVPR.2019.00975.

[52] X. Li et al., "Semantic flow for fast and accurate scene parsing," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K., Aug. 2020, pp. 775–793, doi: 10.1007/978-3-030-58452-8_45.

[53] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiSeNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comp. Vis. (ECCV)*, vol. 11217, Munich, Germany, 2018, pp. 334–349, doi: 10.1007/978-3-030-01261-8_20.

[54] C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, and N. Sang, "BiSeNet v2: Bilateral network with guided aggregation for real-time semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3051–3068, Nov. 2021, doi: 10.1007/s11263-021-01515-2.

[55] S. Kumaar, Y. Lyu, F. Nex, and M. Y. Yang, "CABiNet: Efficient context aggregation network for low-latency semantic segmentation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Xi'an, China, May 2021, pp. 13517–13524, doi: 10.1109/ICRA48506.2021.9560977.

[56] M. Fan et al., "Rethinking BiSeNet for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, USA, Jun. 2021, pp. 9711–9720, doi: 10.1109/CVPR46437.2021.00959.

[57] J. Peng et al., "PP-LiteSeg: A superior real-time semantic segmentation model," 2022, *arXiv:2204.02681*.

[58] H. Pan, Y. Hong, W. Sun, and Y. Jia, "Deep dual-resolution networks for real-time and accurate semantic segmentation of traffic scenes," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3448–3460, Mar. 2023, doi: 10.1109/TITS.2022.3228042.

[59] J. Xu, Z. Xiong, and S. P. Bhattacharyya, "PIDNet: A real-time semantic segmentation network inspired by PID controllers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 19529–19539, doi: 10.1109/CVPR52729.2023.01871.

[60] Z. Li, H. Shen, H. Li, G. Xia, P. Gamba, and L. Zhang, "Multi-feature combined cloud and cloud shadow detection in GaoFen-1 wide field of view imagery," *Remote Sens. Environ.*, vol. 191, pp. 342–358, Mar. 2017, doi: 10.1016/J.RSE.2017.01.026.

[61] S. Mohajerani and P. Saeedi, "Cloud and cloud shadow segmentation for remote sensing imagery via filtered Jaccard loss function and parametric augmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4254–4266, 2021, doi: 10.1109/JSTARS.2021.3070786.

[62] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 1520–1528, doi: 10.1109/ICCV.2015.178.

[63] W. Shi et al., "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1874–1883, doi: 10.1109/CVPR.2016.207.



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